

VIVAN SINHA

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EDUCATION

University of California, Berkeley

Bachelors in Computer Science | Technical GPA: 3.73

Berkeley, CA

Aug. 2021 – May 2025

EXPERIENCE

Scale AI | Full Stack Engineer, Agentic Platform | Python, Go, Terraform, AWS, Helm

July 2025 – Present

- Built core infrastructure for Scale AI's agentic platform, enabling ML and deployment engineers to develop and run customer-facing agents on a shared internal platform
- Developed a Kubernetes-backed control layer to provision isolated sandboxes for agents and MCP servers in under 10 seconds, reducing average startup time by 76% from the previous deployment approach
- Designed and implemented multiple high-priority security features for sandboxed workloads, including configurable network egress controls and a sidecar-based secrets injection mechanism that enabled service access without exposing raw secret values inside agent environments
- Supported large-scale deployments running 50,000+ concurrent sandboxes for production customer workloads

Resolve.ai | Full-Stack Intern | Python, Flask, FastAPI, React, PostgreSQL, Docker

June 2024 – July 2025

- Built a full-stack production application from the ground up using LLMs for ticket analysis
- Developed and maintained backend APIs with Flask, FastAPI, and PostgreSQL and frontend features in React
- Refined the data pipeline with Celery for async task processing and SocketIO for real-time content delivery, reducing report generation time by over 33%
- Automated report generation workflows, reducing required human hours by 84%
- Improved generation accuracy by 27% through model fine-tuning and prompt engineering

OCTO Berkeley | Full-Stack Developer | React, Typescript, Next.js, Vite

September 2024 – Present

- Led development of multiple official student organization websites that serve over 10,000 monthly users

PROJECTS

AI-Powered Staffing App for Oracle | Python, Pandas, OpenAI, Docker

May 2024 – June 2024

- Tasked with creating a lightweight prototype to optimize staffing for thousands of employees
- Advised project-team matching based on a variety of metrics (domain expertise, skill level, experience)
- Reduced staffing time by 87%, allowing managers to spend more time working on projects
- Optimized resource utilization through data-driven matching and cut delivery costs by 22%

Cal Badminton Webpage | Ruby, Jekyll, HTML, CSS, Javascript

October 2023

- Optimized site performance and refreshed the user experience, cutting load time by over 50% and contributing to a 25% increase in applicants the following semester

TECHNICAL SKILLS

Programming Languages: Python, Go, Rust, C++, JavaScript, SQL, Java, C, Ruby

Frameworks & Technologies: Kubernetes, Terraform, Helm, AWS, GCP, Docker, FastAPI, Flask, React, Next.js, Node.js, PostgreSQL, MongoDB, Celery, SocketIO

Developer Tools: Git, GitHub, Bitbucket, Postman, npm

Concepts: Platform Engineering, Infrastructure Automation, CI/CD, REST APIs, WebSockets, Distributed Systems, Scaling, Caching, Sandbox Orchestration, Secrets Management